



Trustworthy analyses of online A/B tests

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Disclaimer

- This will be a “less technical” talk.
- Outline:
 - Introduction;
 - Overview of A/B testing @MSFT;
 - Three (statistical) stories in experimentation.
- Some details are ignored, but references are provided.

Introduction

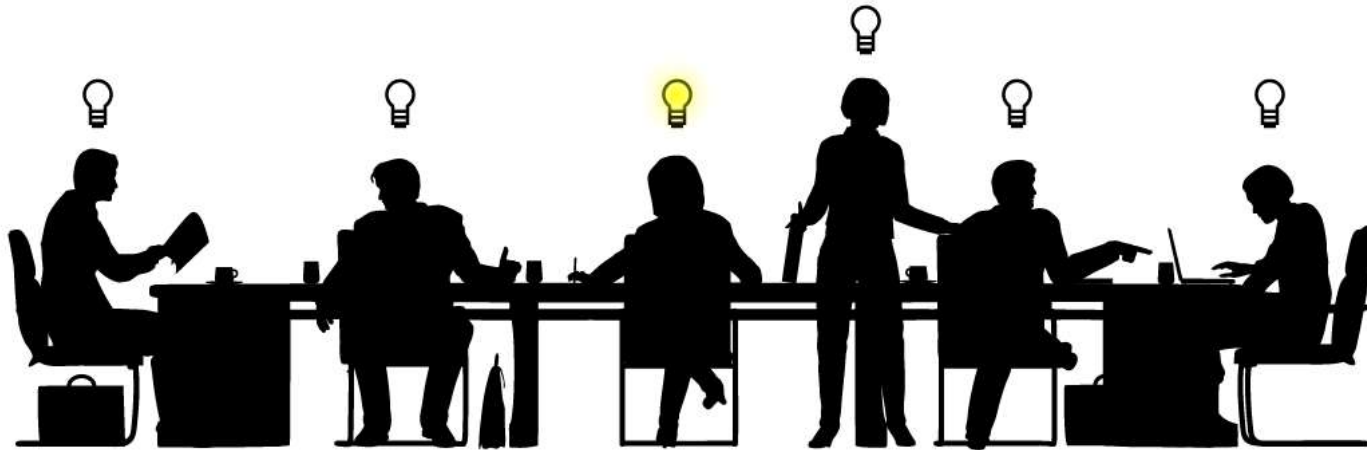
- My [team](#) – making MSFT data-driven, via experimentation:



- [Me](#) – PhD (Harvard, 2015), working as DS/researcher

Overview of A/B testing

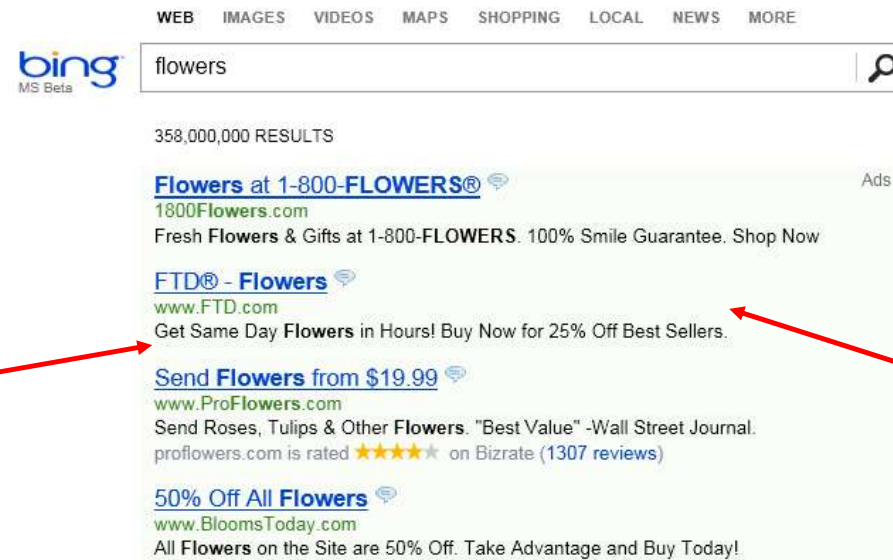
- Most ideas are, frankly, mediocre or terrible.



- Let the (experimental) data speak.

Metric development

- User engagement/satisfaction vs. revenue^[1]:



Gotta make money...

Don't anger the users...

[1] Dmitriev, P. and Wu, X. Measuring metrics. CIKM'16

Metric development

- Quick back rate (QBR):

	Treatment	Control	Delta	Delta %	P-Value
▼ QuickBack					
QuickBack Rate v2				+0.23%	2e-32
--Web Result				+0.79%	≈0
--Answer				+1.48%	≈0
--Ad				-1.10%	≈0

- The trade-off:

Estd Revenue/UU	-2.17%	≈0
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Success criterion

- A valuable experiment:
 - a. Confirm great feature;
 - b. Prevent supposedly great but actually meh feature;
 - c. Prevent bad feature;
 - d. Discover supposedly meh but actually great feature.

Example from Bing

- Longer title for ads^[2]:

Control – existing display

WEB IMAGES VIDEOS MAPS SHOPPING LOCAL NEWS MORE

bing MS Beta

flowers

358,000,000 RESULTS

Flowers at 1-800-FLOWERS® 1800Flowers.com
Fresh Flowers & Gifts at 1-800-FLOWERS. 100% Smile Guarantee. Shop Now

FTD® - Flowers www.FTD.com
Get Same Day Flowers in Hours! Buy Now for 25% Off Best Sellers.

Send Flowers from \$19.99 www.ProFlowers.com
Send Roses, Tulips & Other Flowers. "Best Value" -Wall Street Journal.
proflowers.com is rated ★★★★★ on Bizrate (1307 reviews)

50% Off All Flowers www.BloomsToday.com
All Flowers on the Site are 50% Off. Take Advantage and Buy Today!

Treatment – long titles

WEB IMAGES VIDEOS MAPS SHOPPING LOCAL NEWS MORE

bing MS Beta

flowers

358,000,000 RESULTS

FTD® - Flowers - Get Same Day Flowers in Hours! www.FTD.com
Buy Now for 25% Off Best Sellers.

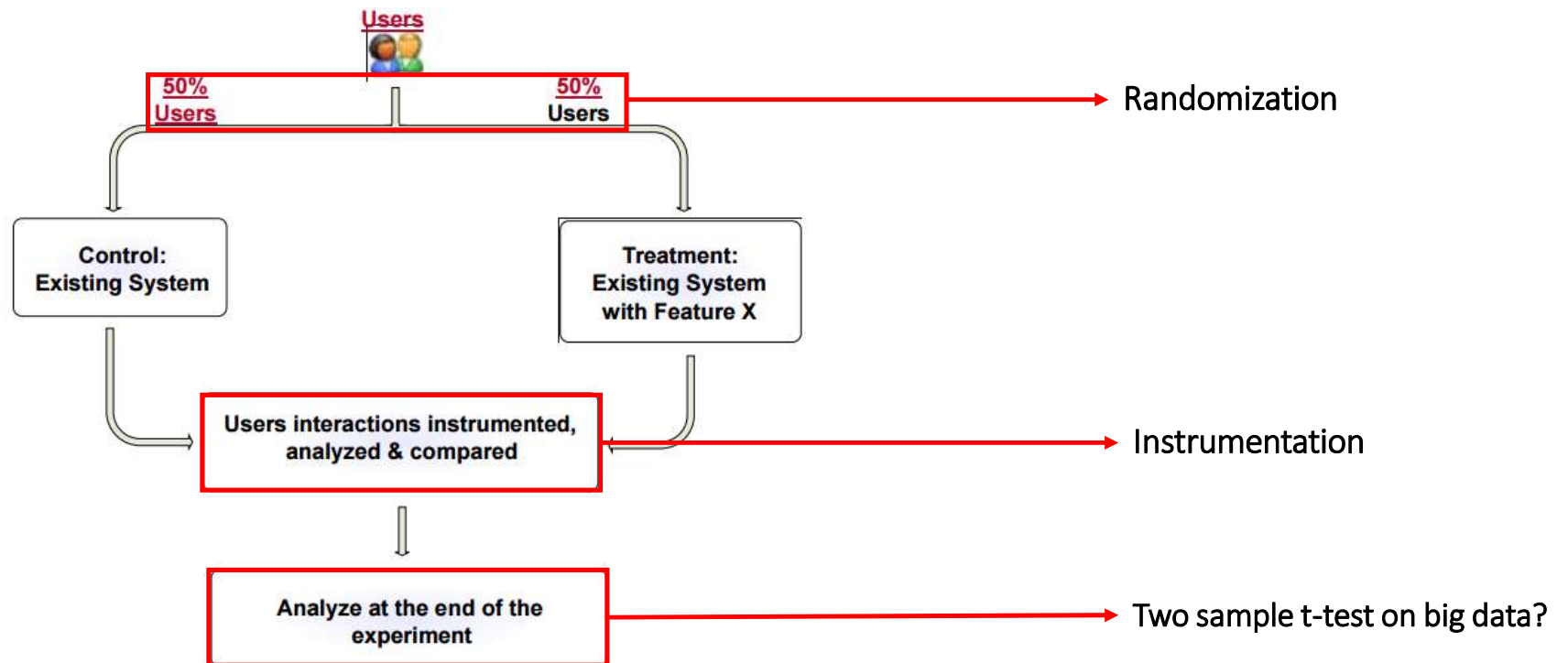
Flowers at 1-800-FLOWERS® | 1800flowers.com 1800Flowers.com
Fresh Flowers & Gifts at 1-800-FLOWERS. 100% Smile Guarantee. Shop Now

Send Flowers from \$19.99 - Send Roses, Tulips & Other Flowers. www.ProFlowers.com
"Best Value" -Wall Street Journal.
proflowers.com is rated ★★★★★ on Bizrate (1307 reviews)

\$19.99 - Cheap Flowers - Delivery Today By A Local Florist! www.FromYouFlowers.com
Shop Now & Save \$5 Instantly.

[2] Deng, A. et al. A/B Testing at Scale: Accelerating Software Innovation. SIGIR'17

Experimentation pipeline^[3]



[3] Kohavi, R. and Longbotham, R. Online Controlled Experiments and A/B Tests. Encyclopedia of Machine Learning and Data Mining. ISBN: 978-1-4899-7502-7.

Story I: Calculating variances

Background

- In Bing, each user views multiple pages;
- Randomization unit (R):
 1. By user: consistent experience;
 2. By page: larger sample size.
- Analysis unit (A):
 1. Page-level metric (business consideration).

Notations

- Number of users n ;
- Randomization probability p ;
- At user-level ($i = 1, \dots, n$):
 - Treatment assignment: $W_{ij} = 1$ if treated;
 - Numbers of treated/control/total calls: N_{iT}, N_{iC}, N_i
 - Observed outcomes: Y_{ij}^{obs} ($j = 1, \dots, N_i$);
 - Sums of treated/control: S_{iT}, S_{iC}
- Estimator:

$$\hat{\tau} = \frac{\sum_{i=1}^n S_{iT}}{\sum_{i=1}^n N_{iT}} - \frac{\sum_{i=1}^n S_{iC}}{\sum_{i=1}^n N_{iC}}$$

Treatment mean $\bar{Y}_T^{\text{obs}}$

Control mean $\bar{Y}_C^{\text{obs}}$

Variance formulas

- R = page – standard method:

$$\widehat{\text{Var}}_S(\widehat{\tau}) = \lambda_T^2 + \lambda_C^2,$$

where

$$\lambda_T^2 = \frac{1}{(\sum_{i=1}^n N_{iT})^2} \sum_{i=1}^n \sum_{j: W_{ij}=1} (Y_{ij}^{\text{obs}} - \bar{Y}_T^{\text{obs}})^2$$

$$\lambda_C^2 = \frac{1}{(\sum_{i=1}^n N_{iC})^2} \sum_{i=1}^n \sum_{j: W_{ij}=0} (Y_{ij}^{\text{obs}} - \bar{Y}_C^{\text{obs}})^2$$

- R = user – Delta method:

$$\widehat{\text{Var}}_D(\widehat{\tau}) = n^{-1}(\xi_T^2 + \xi_C^2),$$

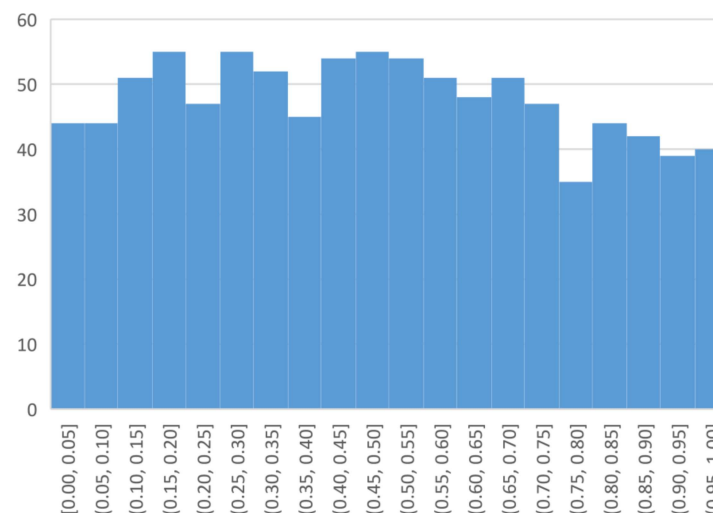
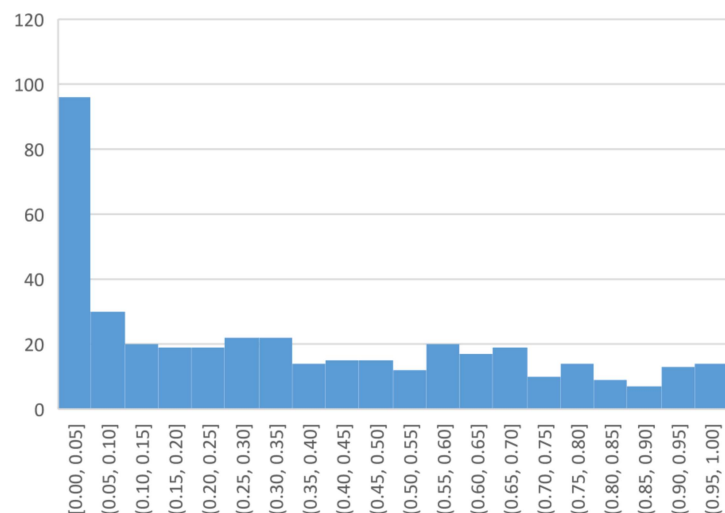
where

$$\xi_T^2 = \frac{1}{(\widehat{\text{E}}N_{iT})^2} \widehat{\text{Var}}(S_{iT}) + \frac{(\widehat{\text{E}}S_{iT})^2}{(\widehat{\text{E}}N_{iT})^4} \widehat{\text{Var}}(N_{iT})$$

$$- 2 \frac{\widehat{\text{E}}S_{iT}}{(\widehat{\text{E}}N_{iT})^3} \widehat{\text{Cov}}(S_{iT}, N_{iT})$$

Results

- P-values for A/A experiments:

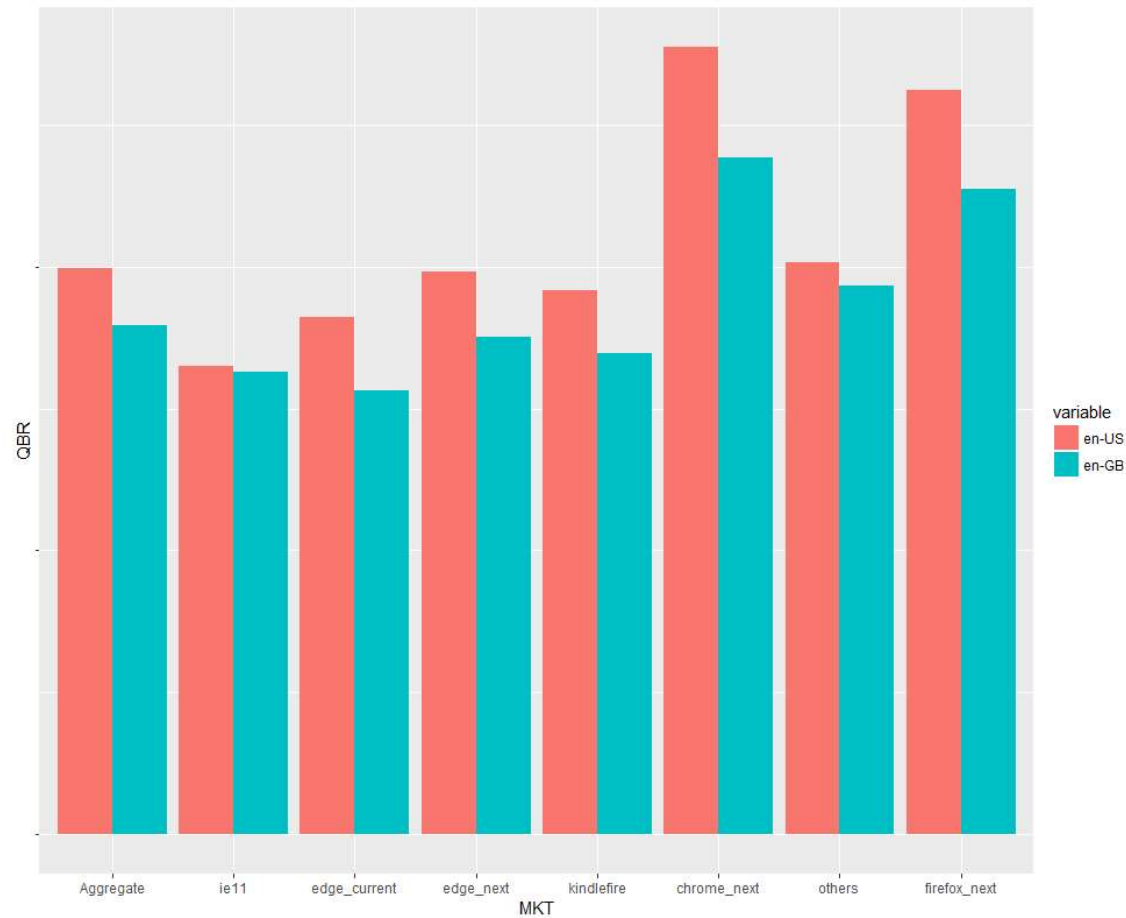


- Actually, there is more – See [Deng, Lu and Litz \(WSDM'17\)](#).

Story II: Finding heterogeneity

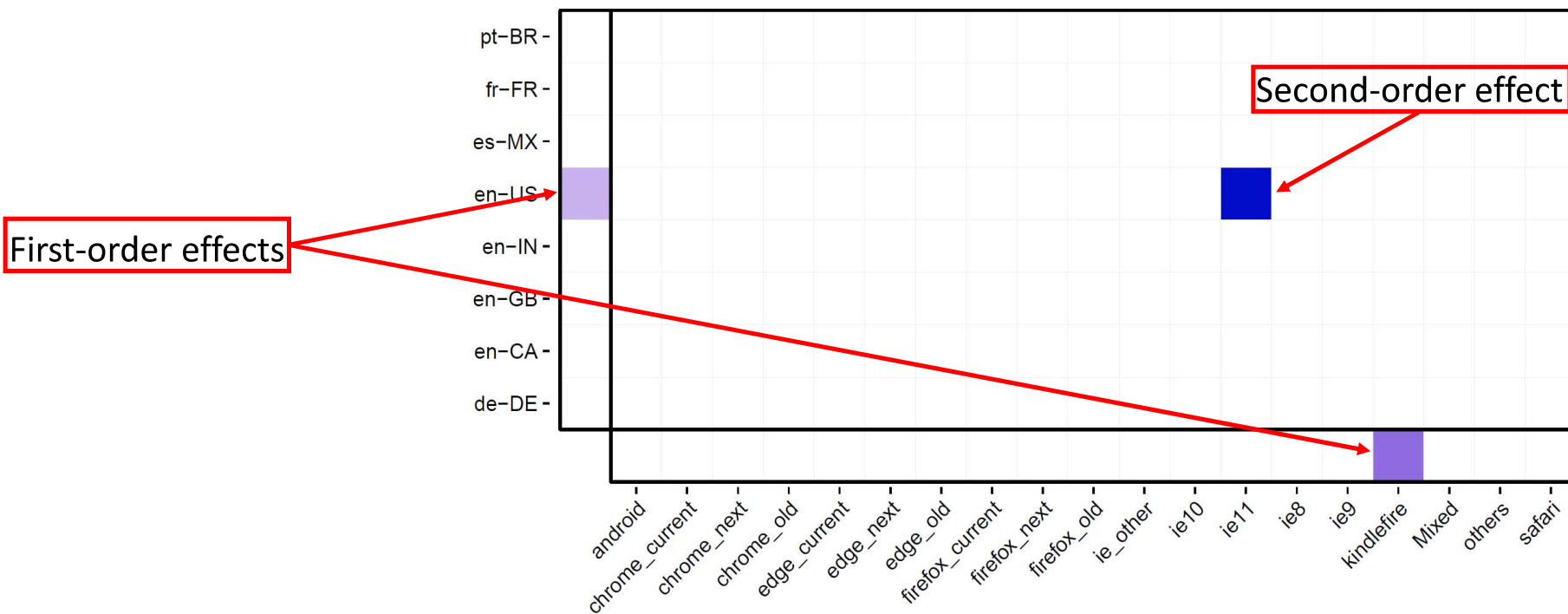
Example

- User behaviors vary across different segments.
- “Personalized” treatment seems necessary.



Results

- A Lasso-type solution^[4]:

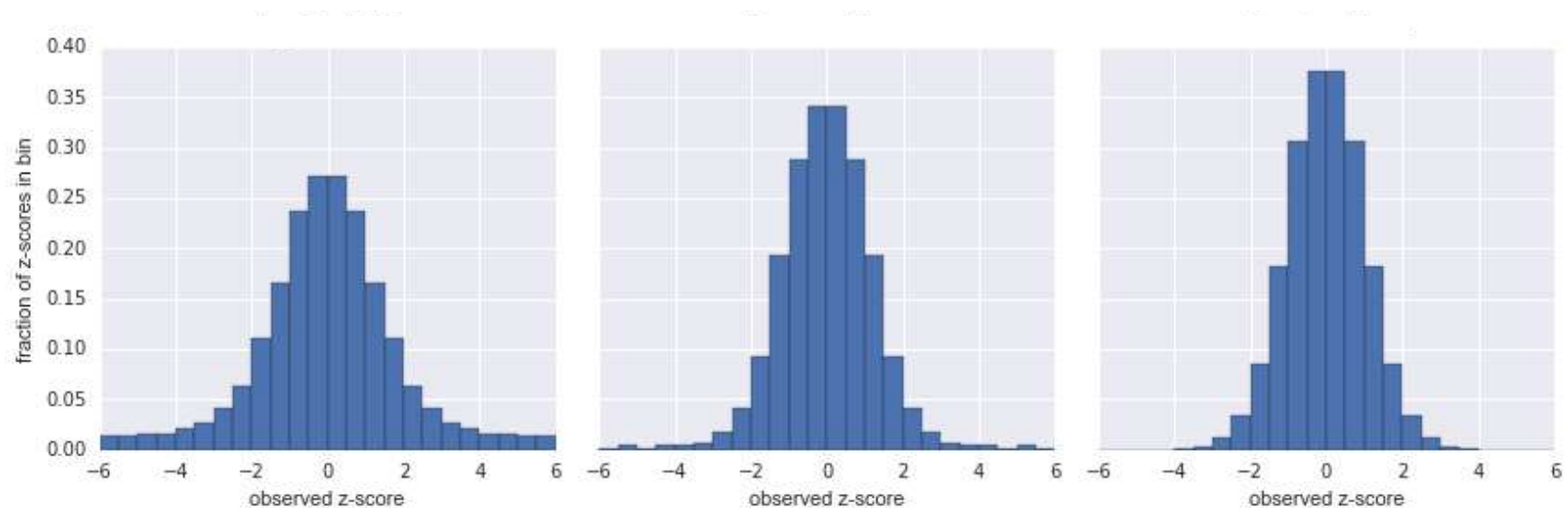


[4] Deng, A. et al. (2016) Concise summarization of heterogeneous treatment effect using total variation regularized regression. ArXiv:1610.03917

Story #3:
(Try to) be Bayesian

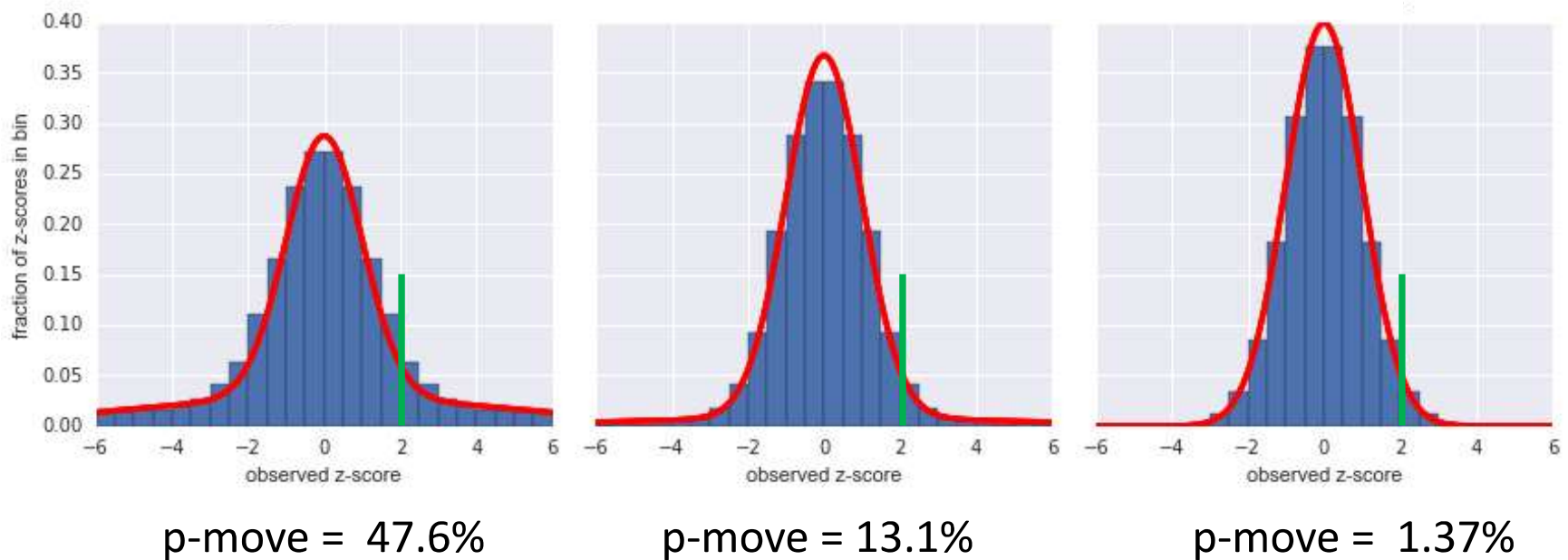
Background

- NOT all metrics are created equal:
 1. Prior information can impact the interpretation of new results;
 2. We want **p-move**: $\Pr(\text{true move} \mid \text{data})$.



Results

- Classic two-group model^[5]:



- Small p-value might NOT indicate real movements.

[5] Efron, B. Microarrays, empirical Bayes and the two-group model. Statistical Science.

Concluding remarks

- Experimentation is at the front-line of technology innovation;
- Trustworthiness is the foundation of experimentation;
- Principled statistical thinking is critical in the age of “Big Data” and “Machine Learning.”